

EDUCATION**Worcester Polytechnic Institute****08/2025 - 05/2027***Master of Science in Data Science – GPA: 4.00***Narsee Monjee Institute of Management Studies****07/2017 – 05/2021***Bachelor of Science in Applied Statistics and Analytics***EXPERIENCE****Worcester Polytechnic Institute - Wildfire Interdisciplinary Research Center****March 2026 - Present****Graduate Research Assistant**

- Developing a probabilistic framework for spatial wildfire risk assessment using Monte Carlo fire simulations (FARSITE), generating burn probability, flame-length, and fireline intensity maps validated against historical California fire perimeters.
- Building geospatial data pipelines integrating LANDFIRE fuels/topography, GridMet weather, HRRR wind fields, and Overture Maps building data to drive fire behavior simulations and HVRA-based spatial risk mapping.

xLM Continuous Validation**Jul 2021 – Jul 2025****Lead Software Engineer - AI**

- Led [Project Manava](#), an AI-driven validation automation platform, from research to production; architected LLM/VLM multi-agent workflows with CrewAI and LangGraph that automated validation lifecycle tasks and reduced manual effort by 90%.
- Built production full-stack features including customer-facing UI, FastAPI services, PostgreSQL/MongoDB data layers, GPU inference endpoints with PyTorch/vLLM, and cloud deployments across AWS/Azure for pharmaceutical validation workflows.
- Implemented CI/CD pipelines with Azure DevOps and Docker, managed Kubernetes/containerized infrastructure, and drafted SOPs for standardized AI-tool usage, version control, release, and validation practices.
- Partnered with product, QA, regulatory, and executive stakeholders to translate regulated business requirements into scalable AI automation products and client-ready demos.

SDET II**Sep 2022 – Oct 2023**

- Designed enterprise automation frameworks with Selenium, Playwright, SpecFlow, and API testing, reducing deployment cycles by 40% through continuous testing and integration.
- Led development of xLMCore NuGet package from v0.0.1 to v2.0.15, creating reusable automation/reporting capabilities across products and teams.
- Built Jira and Azure DevOps integrations improving engineering visibility, workflow traceability, and test execution governance.

SDET I**Jul 2021 – Aug 2022**

- Automated validation testing for enterprise applications including Veeva, Sage, and Rhythm using Selenium, Appium, and Postman while supporting compliance and reliability expectations.
- Established Power BI reporting pipelines for real-time test execution visibility and data-driven QA decisions.

PROJECTS**Few-Shot Driver Verification from GPS Trajectories** | [Repo](#) | *PyTorch, Siamese Networks, ConvTransformer, DDP,**SLURM***Apr 2026**

- Built a Siamese metric-learning model to verify whether two 100-step taxi GPS sub-trajectories came from the same unseen driver; combined multi-scale dilated 1D CNNs, 4-layer pre-norm Transformer blocks, attention pooling, and L_2 -normalized embeddings.
- Trained on 150K balanced trajectory pairs with BCE plus cosine-margin auxiliary loss on $2 \times$ NVIDIA A30 GPUs; achieved 0.9179 validation accuracy and 0.8861 unseen-driver test accuracy, ranking first on the course leaderboard at submission.

Parameter-Golf LLM Ablation Study | [Repo](#) | *PyTorch, Transformer LMs, Quantization, Muon, FineWeb, HPC***Apr****2026**

- Ran 46 controlled 20K-step experiments under a 16 MB model-artifact regime, studying quantization precision, positional encoding, optimizer choice, parameter-efficient layers, multi-query attention, and LR/warmup schedules.
- Found INT8 post-training quantization preserved capacity gains better than INT6 QAT and TurboQuant-INT4; best in-budget model reached 1.2147 bpb at 15.87 MB, while an over-budget 11-layer INT8 model reached 1.1944 bpb at 19.30 MB.

SPGD Optimizer Study on Non-Convex ML Loss Landscapes | [Repo](#) | *PyTorch Optimizers, CIFAR-10, OpenML,**MovieLens, Statistical Evaluation***Apr 2026**

- Implemented Steepest Perturbed Gradient Descent as a PyTorch optimizer and compared SGD, Adam, PGD, RPDG, and SPGD across synthetic functions, OpenML tabular tasks, CIFAR-10/ResNet-18, and MovieLens matrix completion.
- Isolated SPGD's selection-rule effect: SPGD outperformed the random-perturbation control on CIFAR-10 by a mean +3.78 percentage-point paired test-accuracy gap, while Adam remained strongest on well-conditioned landscapes.

GitHub Account Population Estimator | [Repo](#) | *Python, Async APIs, Stratified Sampling, Bootstrap CI, Web Dashboard***Feb 2026**

- Estimated valid GitHub accounts across a 261.7M sequential-ID space using proportional stratified random sampling and the GitHub REST API; built a report-backed web app with async data collection, validation, and dashboarding.
- Queried 16K sampled IDs plus 8K ground-truth validation IDs; estimated 214.1M valid accounts with 95% bootstrap CI [212.5M, 215.7M], 81.8% validity, and 1.48% relative CI width using only 0.009% of the ID space.

Customer Segmentation Analysis | *Python, K-Means, RFM Analysis, Statistical Validation, E-Commerce Analytics* **Dec 2025**

- Clustered Olist e-commerce customers using K-Means after feature engineering 18 behavioral attributes and selecting 7 low-correlation RFM/behavioral features from 74,680 cleaned customer records.
- Identified 6 customer segments with Silhouette score 0.607 and 0.04% test degradation; surfaced an 85% single-purchase rate, review-score differentiation, a small loyal-champion segment, and geographically clustered dissatisfaction.

CATS Anomaly Detection System | *Python, XGBoost, LSTM, Autoencoders, Flask* **Oct 2025 – Dec 2025**

- Built an industrial sensor anomaly-detection system using statistical methods, XGBoost, LSTMs, and autoencoders; deployed inference via Flask API for monitoring workflows.

Sentiment-Based Trading System | *Python, FastAPI, BERT, Selenium* **Feb 2021 – Mar 2021**

- Fine Tuned a BERT model on news based financial news headlines dataset, with technical indicators for intraday trading signals; automated scraping/backtesting and reported 2.6% daily profit with a 1:2.5 risk-reward setup.

TECHNICAL SKILLS

Programming & Systems: Python, C#/.NET, Java, Julia, SQL, Bash, PowerShell, C++, CUDA, Git

Machine Learning & Statistics: PyTorch, Scikit-learn, XGBoost, Pandas, NumPy, PySpark, K-Means, Clustering, Metric Learning, Siamese Networks, Transformers, Time-Series Modeling, Anomaly Detection, Statistical Inference, Bootstrap CIs, Stratified Sampling

Deep Learning & NLP: Transformer Architectures, Attention Mechanisms, Quantization, vLLM, BERT, LSTMs, Autoencoders, Embeddings, Model Evaluation, Distributed Training, Mixed Precision Training

Generative AI & Agents: LLMs, VLMs, RAG, LangChain, LangGraph, CrewAI, DSPy, Multi-Agent Systems, Tool-Calling Agents, AI Workflow Automation

Backend, Cloud & DevOps: FastAPI, Flask, Django, PostgreSQL, MongoDB, SQL Server, Redis, Docker, Kubernetes, Terraform, Azure, AWS, Azure DevOps, CI/CD, REST APIs

Automation & Testing: Selenium, Playwright, Appium, PyTest, SpecFlow, Cucumber, Postman, Test Automation Frameworks, API Testing, Browser Automation

Data Visualization & Analytics: Tableau, Power BI, Matplotlib, Experiment Tracking, Dashboarding, Reporting Automation

Delivery & Compliance: Agile, Scrum, Jira, SDLC, UAT, SOP Drafting, 21 CFR Part 11, GxP Validation

CERTIFICATIONS

Microsoft Azure Fundamentals / Azure DevOps – AZ-900	Jan 2024
Certified Scrum Developer	Aug 2022
Software Development Processes and Methodologies	Jul 2021